



**NATIONAL UNIVERSITY OF MODERN LANGUAGE
ISLAMABAD**

Complex Computing Problem (CCP) Report

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Project Title:

BreatheWell – Lung Disease Detection and Care Assistance

1. Introduction

1.1 Purpose

The purpose of this document is to present a Complex Computing Problem (CCP) analysis based on the Final Year Project BreatheWell – Lung Disease Detection and Care Assistance. This report focuses on requirement gathering and analysis, along with the development of key project management plans including scope management, schedule management, and resource management. The document aims to demonstrate how software engineering principles are applied to manage a complex, AI-based healthcare system.

1.2 Scope

This CCP report covers:

- Requirement gathering and analysis of the BreatheWell system
- Definition and control of project scope
- Planning and management of project schedule
- Identification and allocation of project resources

The report focuses on **project planning and management aspects** rather than full technical implementation.

1.3 Definitions, Acronyms, and Abbreviations

Term	Description
AI	Artificial Intelligence
ML	Machine Learning
COPD	Chronic Obstructive Pulmonary Disease
CCP	Complex Computing Problem
FYP	Final Year Project
UI	User Interface
API	Application Programming Interface
AQI	Air Quality Index

1.4 References

1. Project Proposal and FYP Documentation – *BreatheWell*
2. PMBOK® Guide – Project Management Institute
3. WHO – Respiratory Diseases Overview
4. NUML BS Software Engineering Project Guidelines

1.5 Overview

BreatheWell is an AI-powered web application designed to assist in early detection and management of respiratory diseases using cough and breathing sound analysis. The system integrates machine learning, symptom tracking, environmental alerts, and personalized care recommendations. Due to the involvement of AI, healthcare data, and full-stack development, the project qualifies as a **complex computing problem** requiring structured planning and management.

2. Scope Management Plan

2.1 Scope Definition

The scope of the BreatheWell project includes:

- User registration and authentication
- Symptom submission and health data collection
- Cough and breathing sound recording
- AI-based audio analysis and smart diagnosis
- Personalized care plans and exercise recommendations
- Environmental air quality alerts
- Health history tracking and report generation

2.2 Scope Exclusions

The following are not included in the current project scope:

- Real-time clinical diagnosis or prescription
- Integration with hospital management systems
- Regulatory medical certification

- Physical medical devices integration

2.3 Scope Control

Scope control will be managed by:

- Documenting all requirements clearly
- Avoiding feature additions beyond approved requirements
- Reviewing scope changes with the supervisor
- Updating project documents if changes are approved

2.4 Scope Validation

Scope validation will be performed through:

- Supervisor reviews
- Demonstration of completed modules
- Verification against documented requirements

3. Schedule Management Plan

3.1 Schedule Planning Approach

The **Waterfall Model** is used for project scheduling due to its structured and academic nature. Each phase must be completed before moving to the next.

3.2 Project Phases and Timeline

Phase	Duration
Requirement Analysis	2 Weeks
System Design	2 Weeks
Frontend Development	3 Weeks
Backend Development	3 Weeks
AI Model Development	4 Weeks
Integration	2 Weeks

Phase	Duration
Testing & Evaluation	2 Weeks
Documentation & Final Submission	2 Weeks

3.3 Milestones

- Completion of Requirement Analysis
- Finalization of System Design
- Completion of Core Modules
- AI Model Integration
- Final System Testing
- Project Submission

3.4 Schedule Control

- Weekly progress reviews
- Milestone-based evaluation
- Adjustment of timelines if required with supervisor approval

4. Resource Management Plan

4.1 Human Resources

Team Member	Role
Zulqarnain Hassan	Full-Stack Development, AI Integration
Muhammad Younus	Requirement Engineering, Backend Support
Mumtaz Ali	UI/UX Design, Testing, Documentation
Supervisor	Guidance, Review, Evaluation

4.2 Software Resources

- React.js (Frontend)
- Flask / Node.js (Backend)
- MongoDB (Database)
- Firebase (Authentication)
- Python (AI & ML)
- TensorFlow / PyTorch (AI Models)

- Figma (UI/UX Design)

4.3 Hardware Resources

- Laptop/PC (Core i3 or above)
- Microphone-enabled devices for audio recording
- Internet connectivity

4.4 Resource Allocation Strategy

- Tasks distributed based on individual expertise
- Parallel work on frontend, backend, and documentation
- Efficient utilization of free and open-source tools

4.5 Resource Risk Management

- Backup tools and libraries identified
- Shared repositories for code safety
- Time buffer allocated for technical challenges

Conclusion

This CCP report demonstrates the structured planning and management of the BreatheWell project as a complex computing problem. Through clear requirement analysis and well-defined scope, schedule, and resource management plans, the project is positioned for successful execution and delivery. The application of software engineering and project management principles ensures systematic handling of complexity inherent in AI-based healthcare systems.